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Carbon dioxide reduction from green roofs: A comprehensive review of processes, factors, and quantitative methods

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Highlights

- Analysis of the carbon sequestration and energy savings processes of green roofs.
- Evaluation of factors affecting carbon sequestration and thermal performance.
- Summarizing methods to quantify the carbon dioxide reduction services of green roofs.

Abstract

Green roofs can mitigate climate change through various mechanisms, including carbon

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sequestration, thermal process regulation, and building energy savings. While those mechanisms have been well studied, it remains unclear how to quantify carbon dioxide reduction and design green roofs properly for optimal performance. This review summarizes the processes, factors, and quantitative methods for carbon dioxide reduction from peer-reviewed literature. Firstly, the processes that account for how green roofs reduce carbon dioxide directly (by absorbing ambient CO_2 and storing it in biomass and substrate) and indirectly (thermal process regulation reduces energy consumption) are focused. The factors affecting CO_2 reduction (i.e., vegetation, substrate, building environment, and maintenance) are then discussed to investigate how these factors impact the performance of green roofs. Lastly, quantitative methods, including field experiments and simulation models to evaluate direct and indirect CO_2 reductions from green roofs, are discussed considering different spatial scales. The review found that a better understanding of green roofs' carbon reduction mechanisms is needed. Recommendations also should be developed on customized green roof system designs for various climate conditions. Lastly, there is a lack of green roof decarbonization evaluation at multiple spatial and temporal scales to effectively implement green roofs in urban areas. Overall, this comprehensive review facilitates an understanding of how green roofs reduce CO_2, the driving factors, and quantitative methods, which will advance research on designing and building green roofs as effective climate mitigation solutions in urban areas.

Keywords: Green roof, Urban green infrastructure, Rooftop technology, Carbon sequestration, Energy savings, Climate change mitigation

Word Count: 5925

45

Nomenclature

TERM	*DESCRIPTION*	*UNITS*
AET	Actual evapotranspiration	mm
C_a	Specific heat of air	kJ kg^{-1} K^{-1}
C_s	Specific heat of substrate	kJ kg^{-1} K^{-1}
D	Drainage amount	mm
ET	Evapotranspiration	mm
e_a	Vapor pressure of the air	kPa
e_s	Vapor pressure at the surface	kPa
h	Effective depth of rooting	m
GPP	Gross primary productivity	g C m^{-2} y^{-1}
IR	Amount of irrigation water	mm
K_s	Stress coefficient	dimensionless
k	The extinction coefficient, 0.4 is suggested	
k_c	Crop coefficient	dimensionless
LUE	Light use efficiency	
LW	Longwave radiation	W m^{-2}
$LW_{in/out}$	Incoming/ outgoing longwave radiation	W m^{-2}
NEE	Net ecosystem exchange	g C m^{-2} y^{-1}
NEP	Net primary productivity	g C m^{-2} y^{-1}
PAR	Photosynthetic active radiation	W m^{-2}
$PAR_{bot/top}$	Photosynthetic active radiation beneath/above the canopy	W m^{-2}
PAR_{can}	Photosynthetic active radiation inside canopy	W m^{-2}
P	Precipitation amount	mm
p_c	Readily available water constant	dimensionless
PET	Potential evapotranspiration	mm
PET	Potential evapotranspiration of a specific green roof	mm
Q_{cv}	Sensible heat flux	W m^{-2}
Q_p	Latent heat flux-value	W m^{-2}
Q_s	Heat storage by plants and substrate	W m^{-2}
Q_{tf}	Heat flux transferred into the room	W m^{-2}
Q_{10}	Temperature sensitivity parameter for respiration	
R	R-value, thermal resistance	m K W^{-1}
R_{eco}	Ecosystem respiration	g C m^{-2} y^{-1}
R_n	Net radiation flux	W m^{-2}
$R_{in/out}$	Incoming/outgoing radiation	W m^{-2}
R_{10}	Ecosystem base respiration at 10°C	g C m^{-2} y^{-1}
RAW	Readily available water	mm
r_{ah}	Aerodynamic resistance to heat transfer	
r_{av}	Aerodynamic resistance to vapor transfer	
r_s	Surface resistance, equal to stomatal resistance/ LAI	
SMD	Soil moisture deficit	mm
SW	Short-wave radiation	W m^{-2}
T	Temperature	K
T_a	Ambient air temperature	K
T_s	Surface temperature	K
TAW	Total available water	mm
VPD	Vapor pressure deficit	Pa
VWC	Volumetric water content	mm
z	Substrate depth	m

Greek letters

α	Albedo	dimensionless
γ	Psychrometric constant	kPa K^{-1}
θ	Moisture content	%
ε	Emissivity of the surface	dimensionless
σ	Stefan-Boltzmann constant	$5.67 \times 10^{-8} W m^{-2} K^{-4}$
σ_f	Fractional vegetation coverage	%
Λ	Latent heat of vaporization at ground temperature	J kg^{-1}
ρ_a	Mean air density at constant pressure	kg m^{-3}
θ_{fc}	Water content at field capacity	%
θ_{wp}	Water content at wilting point	%
λ	Thermal conductivity of substrate	W m^{-1} K^{-1}

46

1. Introduction

Rapid urban expansion has increased energy demand and greenhouse gas (GHG) emissions, significantly contributing to climate change. By 2050, 8.5-9.9 billion people will live in cities, accounting for 55-78% of the world's population [1, 2]. Buildings are the largest consumers of energy, releasing about 40% of the world's total GHG emissions [3]. Therefore, building design that reduces CO_2 emissions and improves energy efficiency is an important direction in the building sector to meet low-carbon goals in the face of climate change.

Various rooftop technologies have been proposed as potential climate mitigation and adaptation solutions because roofs generally account for about 20-25% of the total urban surface area [4]. Green roofs (GRs), or vegetated roofs, have been considered an effective Nature-based Solution and have been adopted worldwide to improve urban sustainability and resilience [5]. The significant reductions in energy consumption, CO_2 emissions, and waste production associated with GRs render them a cleaner environmental option. The scientific literature has extensively verified that GRs sequester carbon (CO_2) and improve building energy performance – both of which contribute to reducing carbon dioxide and mitigating climate change [6-8].

Carbon dioxide can be reduced directly and indirectly through GRs. Ambient CO_2 is directly absorbed by vegetation from the atmosphere and stored in biomass and substrate (direct reduction), while thermal regulation reduces building energy consumption (indirect reduction). However, the reduction potential of GRs to mitigate climate change is often poorly quantified, and the mitigation potential is underestimated because many interacting processes and mechanisms go into the quantification.

Previous studies have mainly focused on field experiments, simulation tools, or both to

analyze GR processes [9]. Controlled and/or long-term field experiments quantify GR performance under different meteorological conditions and/or by altering vegetation and substrate properties [10]. In field experiments, net ecosystem primary productivity (*NEP*), meteorological parameters, conductive heat fluxes, and heat load for cooling/heating can be measured directly. In contrast, GR simulations can evaluate carbon-related and thermal properties of GRs at different temporal and spatial scales and for various "what-if" scenarios [11, 12]. Several recent studies have summarized and reviewed the environmental benefits and ecosystem services of GRs [9, 13]. Various key mechanisms (e.g., carbon sequestration, heat and mass transfer, and energy consumption reduction) of GRs are associated with carbon dioxide reduction. Although reviews on these key mechanisms have been published individually [6, 8, 14, 15], they have not been synthesized comprehensively to quantify the climate change mitigation potential of GRs.

This study aims to provide a comprehensive analysis of the related processes, factors, and methods to quantify the carbon dioxide reduction of GRs. The present work advances this goal by analyzing direct and indirect carbon-reduction processes (Section 2) and identifying and summarizing the factors and properties of GRs that influence the related physical and biophysical processes (Section 3). Finally, methods to quantify the processes and evaluate the carbon-reduction benefits are presented (Section 4). By analyzing the processes by which GRs sequester carbon and mitigate thermal stress, and by analyzing the main factors and summarizing the methods how previous studies assessed the decarbonization of GRs, this study will advance research on how to design and build GRs as climate mitigation and adaption solution in urban areas.

2. Processes of green roof carbon dioxide reduction

A typical GR configuration comprises at least three layers: vegetation, substrate, and waterproof membrane. Between the waterproof membrane and substrate, some GRs also include a root repellant, a filter membrane, and a drainage and water storage layer [14], as shown in Fig.1.

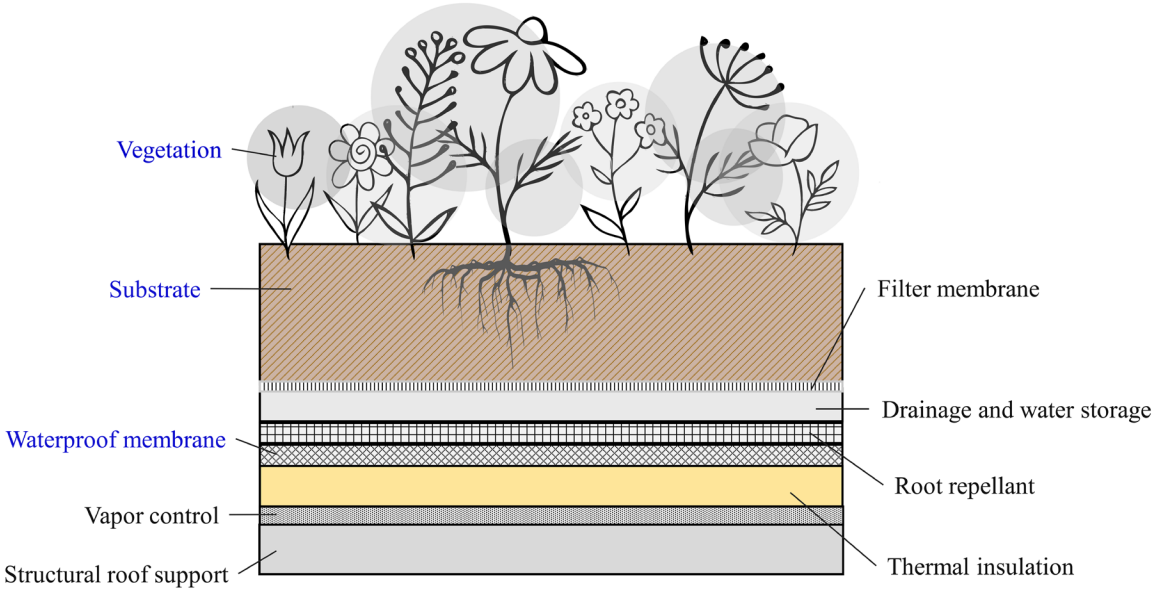


Figure 1. Schematic structure of a typical green roof.

In two steps, this section establishes the relationship between GR effects on carbon dioxide reduction and associated processes. Firstly, the carbon dioxide reduction benefits of GRs and associated processes generated by retrofitting GRs were connected to provide an overview of carbon dioxide reduction processes in section 2.1. Then, each process in terms of carbon cycling and thermal process regulation was analyzed to understand the mechanisms of how GRs reduce carbon dioxide directly and indirectly in sections 2.2 and 2.3, respectively.

2.1 Overview of carbon dioxide reduction processes

GRs can reduce CO_2 directly and indirectly [6] (Fig. 2). GR vegetation directly absorbs

ambient CO_2 from the atmosphere through photosynthesis, which then is stored in plants and substrates as above and below-ground biomass and substrate organic matter (Fig. 2A) [16]. Vegetation and substrate above rooftops can be used as a passive technique to mitigate air/surface temperatures and conductive heat fluxes, thus reducing cooling energy demand and lower CO_2 emissions (Fig. 2B). In addition, GRs reduce temperature fluctuations of the roof membrane to extend its life span and reduce carbon released during manufacturing [17]. GRs can increase carbon sequestration, improve ambient thermal performance, and decrease indoor energy demand to directly and indirectly reduce carbon dioxide (Table 1).

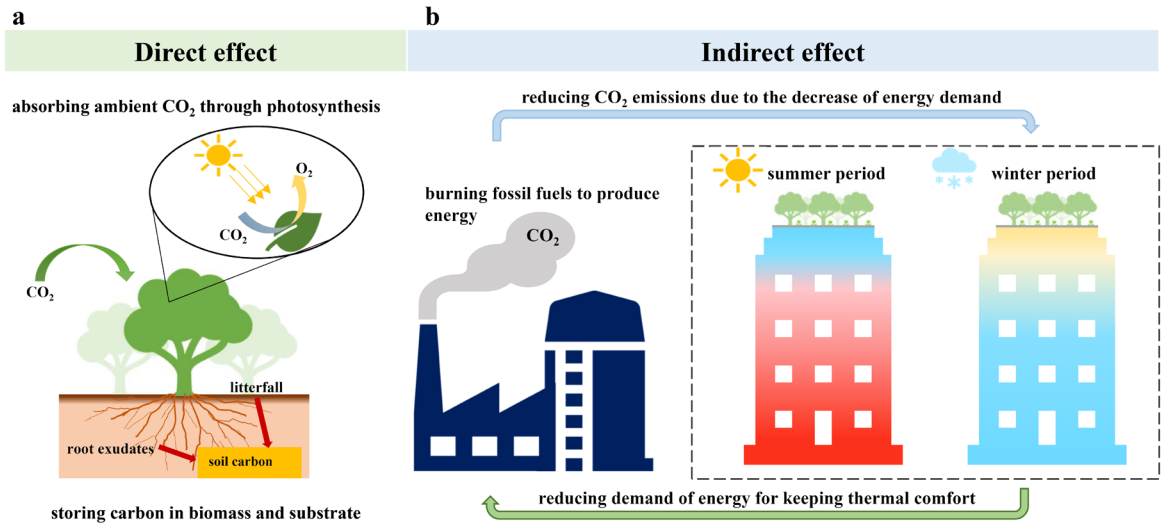


Figure 2. Conceptual graphic of direct (A) and indirect (B) effect of green roofs on carbon dioxide reduction.

The direct and indirect effects correspond to several associated benefits that performance metrics can quantify. As shown in Table 1, the benefits associated with direct effects include air CO_2 absorption and carbon storage, which are evaluated by ambient CO_2 concentration around the canopy and carbon amount on plants and soils, respectively. On the other hand, the benefits related to the indirect effects are based on urban cooling and reduced

indoor energy demand, which are assessed by the Bowen ratio (Q_{cv}/Q_p), and heating/cooling load, respectively. Additionally, the relevant processes that significantly impact the performance metrics in Table 1 include net ecosystem exchange (or CO_2 uptake), net radiation flux, sensible heat flux, latent heat flux, shading effect, thermal mass, and insulation effect.

Table 1

Important processes and properties (rows) and carbon-related benefits (columns) that they influence. The [+/-] symbol refers to the direction of change needed in the metric for improvement. Markers represent a change in direction (● positive; ● negative). Notably, heating or cooling loads are dependent on meteorological conditions. In most cities, heating and cooling periods both occur within a year.

Benefits→	Direct impacts		Indirect impacts		
	Air CO_2 absorption	Carbon storage	Ambient thermal mitigation	Indoor energy demanding	
Evaluation metrics→ Processes ↓	Ambient CO_2 concentration [-]	Carbon amount on plants and soils [+]	Bowen ratio [-]	Heating load [-]	Cooling load [-]
Net primary productivity (or CO_2 uptake)	●	●			
Net radiation flux	●	●		●	●
Sensible heat flux			●		
Latent heat flux			●	●	●
Shading effect				●	●
Thermal mass				●	●
Insulation				●	●

2.2 Process of direct carbon dioxide reduction

GRs act as carbon sinks or sources [18]. GRs are an innovative practice to capture and store the environment's existing CO_2 [19]. Fig. 3 shows the carbon cycle of the GR ecosystem. There are five carbon pools related to the GR ecosystem, from top to bottom, including the atmospheric pool, above-biomass, litter layer, below-biomass, and substrate pool (Fig.3. A~

E).

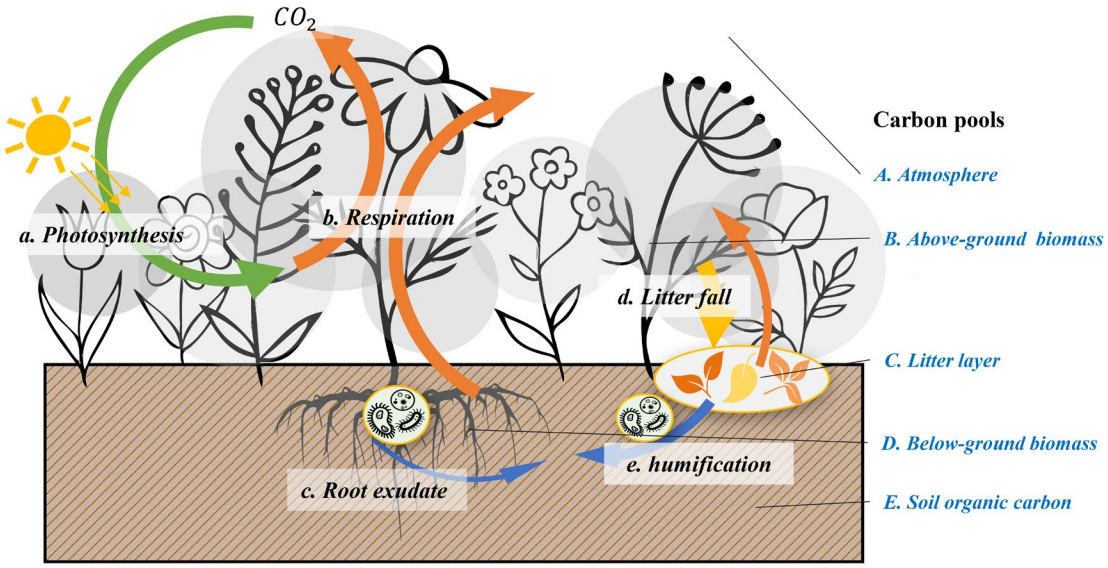


Figure 3. Conceptual graphic depicting the carbon cycling of green roof ecosystems. Capital letters denote the carbon pools: (A) atmosphere, (B) above-biomass, (C) litter layer, (D) below-biomass, and (E) substrate pool. Small letters refer to processes related to the conversion of carbon forms: (a) Photosynthesis, (b) Respiration, (c) Root exudate, (d) Litterfall, (e) humification.

The exchange of CO_2 between GRs and the overlying atmosphere happens through photosynthesis (Fig. 3. a) and respiration (Fig. 3. b). As the primary input of carbon cycling, CO_2 is absorbed through photosynthesis by the GR. CO_2 fluxes are released into the atmosphere by autotrophic respiration and heterotrophic respiration. Net primary productivity (NEP) is the net CO_2 uptake. The gross flux of Gross primary productivity (GPP) into the ecosystem and ecosystem respiration (R_{eco}) out of the land system are the two opposing parts of NEP. The ability of GRs to sequestrate CO_2 can be depicted by NEP.

$$NEP = GPP - R_{eco} \quad (1)$$

Acting as the medium that CO_2 fluxes flow into the GRs, plants can combine CO_2 and water to synthesize organic compounds (mainly carbohydrates) called photosynthesis [20].

Energy for photosynthesis is derived from solar radiation. Therefore, the CO_2 assimilation can be modelled by the parameterization of photosynthetically active radiation (PAR) (Table 2 Eq. 2-4) [21]. GPP is estimated by a light use efficiency (LUE) model (Eq. 5-6) [22]. R_{eco} is the combination of autotrophic respiration and heterotrophic respiration (leaves, stems, roots, microorganism, and soil animals etc.). As an enzyme-mediated biological reaction, respiration is dependent on temperature. A general approach to model R_{eco} is using temperature as the dominant driver (Eq. 7) [23].

Table 2

Related equations for calculating NEP.

Terms	Description	Equation		Ref.
PAR_{can}	PAR_{can} can be estimated by the meaning PAR value from top to bottom of the canopy. The PAR inside the canopy is reduced according to the Beer-Lambert law.	$PAR = 0.5 \times SW_{in}$	(2)	[21]
		$PAR_{bot} = PAR_{top} e^{-kLAI}$	(3)	[21]
		$PAR_{can} = \frac{PAR_{top} + PAR_{bot}}{2}$	(4)	[24]
GPP	The energy needed for photosynthesis comes from solar radiation, thus, CO_2 assimilation could be modelled by the parameterization of PAR.	$GPP = PAR_{can} \times LUE$	(5)	[22]
LUE	The value of LUE depends on biological factors (e.g., plant type, vegetation structure, and stomatal resistance) and abiotic factors (e.g., water availability, radiation, temperature, nutrient, and ambient CO_2 concentration).	$LUE = LUE_{\max} \times f(T, VWC, ...)$	(6)	[22]
R_{eco}	As an enzyme-mediated biological reaction, respiration is dependent on temperature.	$R_{eco} = R_{10} Q_{10}^{\frac{T-10}{10}}$	(7)	[23]

Carbon cycling within the GR ecosystem is complex. Carbon in the forms of CO_2, organisms, dead tissue, and soil/substrate organic carbon (SOC) could transform from one pool to another through different processes, e.g., photosynthesis, respiration, root exudate (Fig.3. c), litterfall, decomposition, and humification. The amount of litter (Fig. 3. d) depends

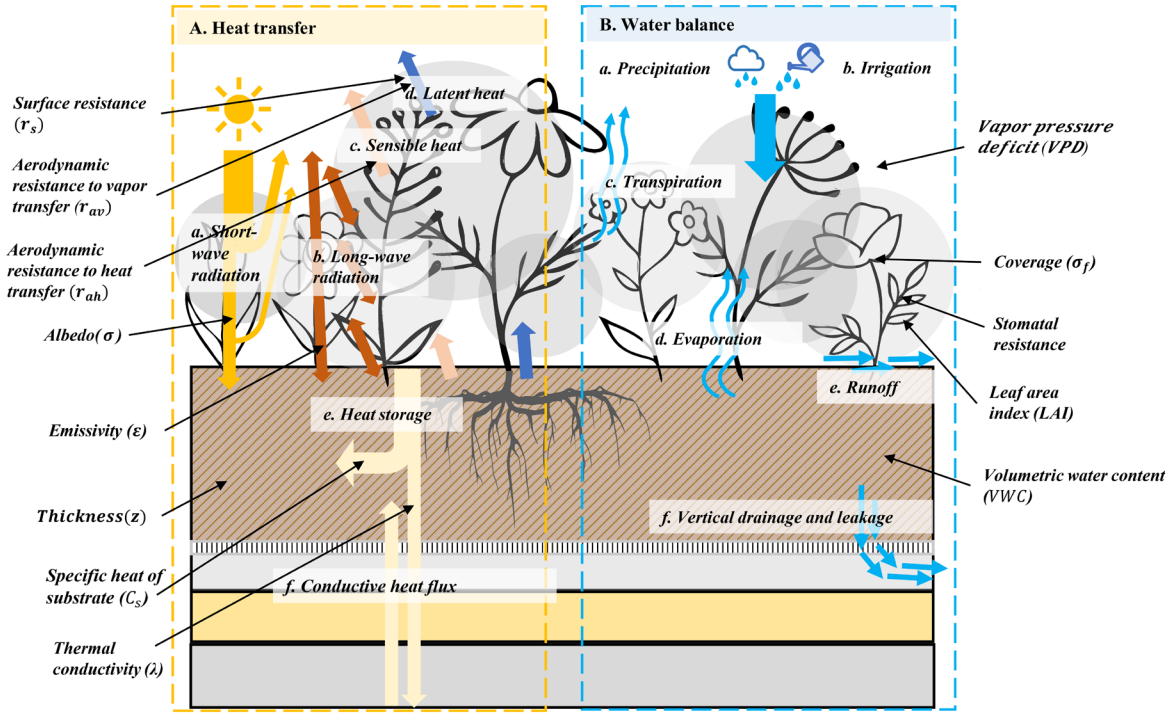
on plant phenology and water and nutrient availability [25]. Decomposition is the key ecological process through which litter decays with rainwater leaching, insect activity, and microbial activity [26]. The decaying organic would release large amounts of CO_2 into the atmosphere, and a small fraction would be transferred to humic substances (Fig.3. e), which can be stored in the soil for a long time [27]. The decomposition rate is controlled by litter quality, temperature, rainfall, and humidity [28].

2.3 Process of indirect carbon dioxide reduction

2.3.1 Heat transfer

The indirect carbon dioxide reduction effect mainly stems from the thermal process regulation of GRs. Heat transfer through GRs differs from traditional bare roofs due to the atmosphere-GRs heat interaction, shading effect, thermal mass, and insulation effect from GRs. The heat fluxes on GRs are shown in Fig. 4A. Substrates and plants absorb the incident (global) solar radiation reaching the surface; parts of it will be reflected. (Fig.4A. a). Longwave radiation is emitted by the surface (Fig.4A. b), and the sensible heat exchange by convection (Fig.4A. c) and the latent heat flux from evapotranspiration (Fig.4A. d) also act as pathways to lose received heat of the GRs. Part of the heat is stored (Fig.4A. e) or/and transferred through the roof into the room (Fig 4A. f). Plants and substrates exchange energy with the roof, and the thermal environment can be depicted as Eq. (8) (see Table 3):

$$R_n = Q_{cv} + Q_p + Q_s + Q_{tf} \quad (8)$$



188

189 **Figure 4.** (A) Simplified graphic for energy transfer in green roofs in the left blue-dotted line frame.
190 (a) Short wave radiation, (b) long-wave radiation, (c) sensible heat, (d) latent heat, (e) heat storage,
191 (f) conductive heat flux. (B) water balance for the substrate of GRs in the yellow-dotted line right
192 frame. (a) precipitation, (b) irrigation, (c) transpiration, (d) evaporation, (e) runoff, (f) vertical
193 drainage and leakage.

194 **Table 3**

195 Calculation methods for the terms in Eq. (8). The equations of the terms have been well
196 summarized by Cook and Larsen [13].

Terms		Equations		Ref.
R_n	Net radiation flux	$R_n = R_{in} - R_{out} = SW_n + LW_n = SW_{in}(1-\alpha) - \varepsilon\sigma T_s^4 + \varepsilon LW_{in}$	(9)	[29]
Q_{cv}	Sensible heat flux	$Q_{cv} = \frac{\rho_a C_a}{r_{ah}}(T_s - T_a)$	(10)	[30]
Q_p	Latent heat flux	$Q_p = \Lambda ET = \frac{\rho_a C_a}{\gamma(r_s + r_{av})}(e_s - e_a)$	(11)	[31]
Q_s	Heat storage by plants and substrates	$\Delta Q_s = C_s \frac{dT}{dt} dz$	(12)	[32]
Q_{tf}	Heat transferred reaching building envelope	$Q_{tf} = \lambda \frac{dT}{dz}$	(13)	[29]

2.3.2 Atmosphere-GRs heat interaction

Interactions of GRs with the atmosphere occur mainly through radiative heat fluxes, sensible heat fluxes, and latent heat fluxes. Radiative heat fluxes from the Sun determine the energy transfer of GRs [33]. The net radiation flux (R_n) refers to the difference between incoming and outgoing radiation ($R_{in} - R_{out}$), or the combination of net shortwave and longwave ($SW_n + LW_n$) (Table 3- Eq. 9). The SW radiation gain is controlled by the albedo (α) of the GR surface. The net longwave radiation (LW_n) emitted to the atmosphere is dependent on the surface temperature (T_s), emissivity (ε), and LW_{in}. About 20% of solar radiation is reflected back into the atmosphere on a GR [34]. Furthermore, GRs cool the environment by partitioning more available energy into latent heat than sensible heat. The Bowen ratio (Q_{cv} / Q_p), which denotes the partition ratio of the sensible and latent heat flux, is often used to evaluate the cooling ability of green infrastructure [35]. The sensible heat flux (Q_{cv}) depends on the temperature difference between surfaces and the air, wind speed, and aerodynamic resistance to heat transfer (r_{ah}) (Table 3-Eq. 10). The latent heat flux (Q_p) depends on surface resistance (r_s) and aerodynamic resistance to vapor transfer (r_{av}), wind speed, air temperature, vapor pressure deficit (VPD), and water availability (Table 3 -Eq. 11).

2.3.4 Shading effect

The cooling benefits from shade have been observed for ground-based green infrastructure [36] and GRs [37]. Shade refers to the attenuation of radiative fluxes obstructed by the canopy layer [38]. Applying the Beer-Lambert law establishes a method to simplify the radiative fluxes through the canopy (Table 2 -Eq. 3) [24, 39]. Shade reduces the net radiative heat incident on a surface, decreasing surface temperature and heat storage [40].

GRs can also reduce air temperature near the canopy [41]. For example, [37] observed lower air temperature below the tree canopy than above during the summer.

2.3.5 Thermal mass and insulation effect

The conductive heat flux of GRs is lower than that of traditional roofs if the specific heat capacity (C_s) of the roof's substrate is large [42]. The substrate layer of GRs can be considered a large heat sink on rooftops, and a considerable amount of energy can be stored without increasing the soil temperature [37]. In addition, the conductive heat flux that reaches the building envelope through the GR is more stable due to the higher thermal mass [43]. The difference between the highest and lowest diurnal surface temperature is usually lower than that of a traditional bare roof [44].

The insulation effect of GRs slows the heat transfer in or out of a building. The R-value describes the insulation property of buildings as a function of the temperature difference and material thickness (Eq. 14):

$$R = \frac{\Delta T}{Q_{tf}} \tag{14}$$

While excess heat of a building roof is transferred to the top-story rooms when it is hot, the conductive heat flux (Q_{tf}) is reversed in the cold season when rooms are heated (Table 3-Eq. 13). GRs considerably attenuate and delay conductive heat fluxes [45]. Building energy savings from GRs are usually larger for buildings without insulation [45]. Jim and Tsang [37] found that a substrate thickness of 10 cm is sufficient to reduce the heat penetration into the building considerably. Simulations confirmed that the thermal load is reduced with increasing insulation [46].

240 *2.3.6 Water balance*

241 The carbon uptake of plants has a tight coupling with water availability because they
242 close stomata in response to water stress to avoid hydraulic failure [47]. GRs mitigate urban
243 heat stress through evapotranspiration (*ET*) [31].

244 In addition, the substrate's volumetric water content (*VWC*) regulates the conductive
245 heat transfer process. Therefore, the water balance of GRs plays an important role in both
246 direct and indirect carbon dioxide reduction.

247 Solar energy is partitioned into latent rather than sensible heat. The latent heat flux (Q_p)
248 can be calculated from *ET* and the latent heat of vaporization (Λ) (Table 4- Eq. 11). As the
249 substrate is considered a horizontally-uniform non-deformable porous medium, the
250 substrate's volumetric water content depends on the water budget (Fig. 4B). The water input
251 includes precipitation (Fig. 4B. a) and irrigation (Fig. 4B. b). The output includes *ET* (Fig 4B
252 c, d) and drainage (Fig. 4B. e, f). This process is described by the GR water balance model
253 [48, 49] and expressed as follows:
254

$$\Delta VWC = IR + P - ET - D \quad (15)$$

255 *ET* can be calculated using the Penman-Monteith equation (Table 4 -Eq. 16) [50]. More
256 recently, Cuthbert, Rau [51] proposed a soil moisture balance model (SMBM) to simulate
257 *ET*. The SMBM calculates the total available water (*TAW*) as the difference between the water
258 content at field capacity (θ_{fc}) and wilting point (θ_{wp}) multiplied by the effective depth of
259 rooting(h) (Table 4 -Eq. 17). *TAW* in the substrate can limit the *ET* ability of GRs. The plant
260 transpiration and substrate evaporation were integrated into the crop coefficient (k_c) to
261 estimate potential evapotranspiration (PET_s) of a specific GR (Table 4 -Eq. 19). When the

soil moisture deficit exceeded the readily available water (RAW), the stress coefficient (K_s) and the actual evapotranspiration (AET) were assumed to decrease linearly (Table 4 -Eq. 18-20). Coupling models with mass balance calculations can make energy balance simulations more accurate [11], and the thermal performance of GRs can then be explored under different irrigation and precipitation scenarios [52, 53].

Table 4

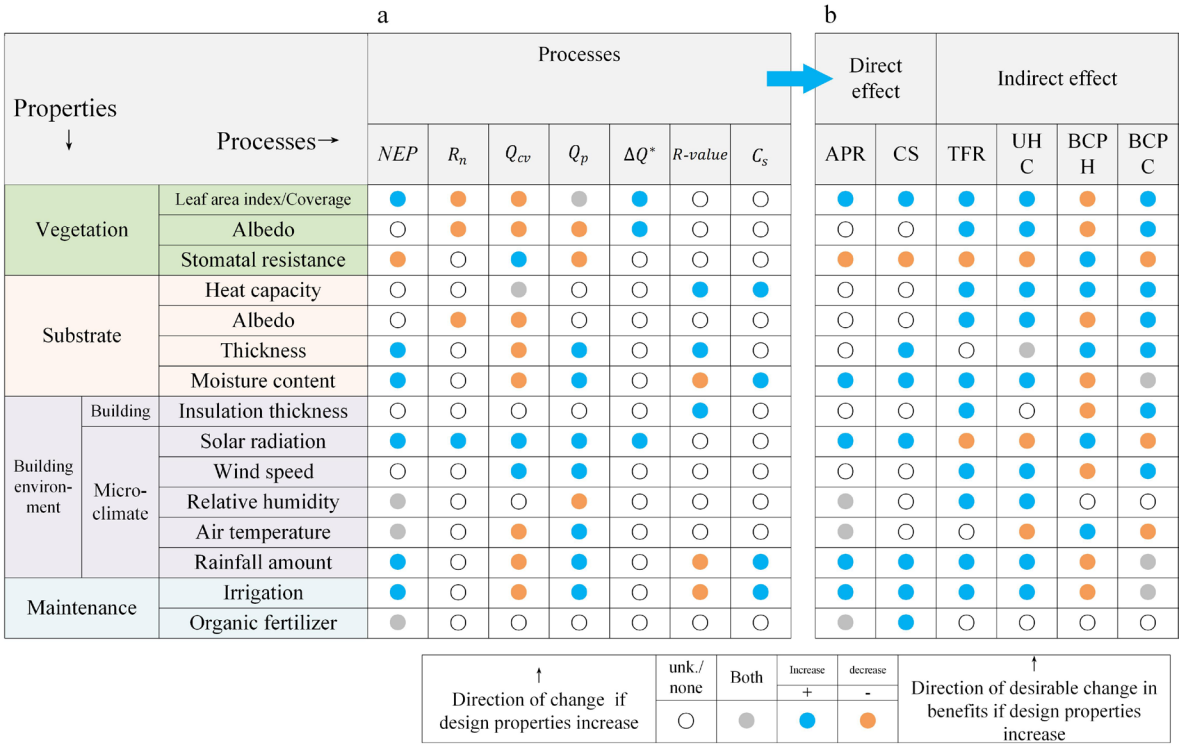
Equations to estimate evapotranspiration using the soil moisture balance model (SMBM) proposed by Cuthbert, Rau [51].

Terms		Equations		Ref.
PET	Potential evapotranspiration	$PET = \frac{0.408\Delta(R_n - G) + \gamma \frac{900}{T+273} u_2 (e_s - e_a)}{\Delta + \gamma(1 + 0.34u_2)}$	(16)	[50]
TAW	The total available water	$TAW = h(\theta_{fc} - \theta_{wp})$	(17)	[51]
RAW	Readily available water	$RAW = p_c TAW$	(18)	[51]
AET	Actual evapotranspiration	$AET = K_s PET_s = K_s k_c PET$	(19)	[54]
K_s	Stress coefficient	$K_s = \frac{TAW - SMD}{TAW - RAW}$	(20)	[51]

3. Factors related to carbon dioxide reduction

This section outlines the design factors of a green roof that contribute to the processes covered in Section 2. Peer-reviewed literature is summarized to elicit how different design and maintenance factors influence the physical processes and environmental benefits of GRs. The literature is divided primarily between GR field experiments conducted to investigate direct carbon dioxide reductions and simulation experiments to evaluate the thermal and energy performance of GRs [8]. Fig. 5 presents how the design parameters related to vegetation, substrate, building environment, and maintenance discussed in this section

279 influence the physical processes (Fig.5A) and carbon dioxide reduction (Fig.5B).



280

281 **Figure 5.** (A) Direction of change in values of the physical processes if the design parameters
282 (factors) are increased. (B) Direction of desirable change in benefits if the design parameters are
283 increased. Note: ACA =Air CO_2 absorption, CS =Carbon storage, ATM= Ambient thermal mitigation,
284 and IED = Indoor energy demanding (IED-heating = Indoor energy demanding during heating periods,
285 IED-cooling = Indoor energy demanding during cooling periods). Blue represents an increase; orange
286 represents a decrease.

287

288 *3.1 Vegetation*

289 Vegetation is the top layer and the most essential component of GRs. Vegetation
290 interacts with the atmosphere through shading, photosynthesis, respiration, and transpiration.
291 As shown in Fig. 5, vegetation parameters that significantly influence GR performance are
292 structural properties (e.g., leaf area index, plant height, coverage), spectral properties

(albedo), and stomatal resistance [55, 56].

3.1.1 Structural properties (*LAI/ Coverage*)

The Leaf area index (*LAI*) refers to the projected area of leaves over a unit of land and ranges between 0.5~5.0 [56]. *LAI* is the most relevant vegetation parameter affecting the carbon dioxide reduction of GRs [55] because it is directly related to photosynthesis, shading, evapotranspiration, and convective heat transfer. In addition, the ambient CO_2 absorption rate increases with *LAI* (Table 2 Eq. 2-5). The canopy growth rate also impacts carbon dioxide reduction because the canopy intercepts incoming radiation [57]. Less solar radiation reaches the roof surface as *LAI* increases (Table 2 -Eq. 3). Yaghoobian and Srebric [58] found that the daily radiative flux reaching an unplanted control surface was 32% (6.2 $kWhm^{-2}$) higher than that of a shaded surface from the dense canopy. They estimated that the annual building cooling load decreased by 13 $kWhm^{-2}$, while the heating load increased by 0.88 $kWhm^{-2}$ due to the GR. Another experiment showed that an increase in *LAI* from 2 to 5 reduces the heat transmitted to the soil by 250W m^{-2} [59]. Lundholm, Tran [60] reported that the water loss rates through *ET* of surfaces covered with *Sedum spurium* were ~16% lower than that of bare soil. Similar results were found for other species (*Danthonia spicate, Empetrum nigrum, and Vaccinium macrocarpon*). Lower cooling and higher heating load were observed with higher coverage.

3.1.2 Spectral properties (*Albedo*)

Spectral reflectivity (albedo) is the ratio of reflected radiation to incoming solar radiation. The plant albedo plays an important role in net radiation. Experiments confirmed that a semi-GR (0.16) had higher albedo than an extensive GR (0.12) [43]. Zhao, Tabares-Velasco [61] simulated two types of plant-substrate assemblies with high and low reflectivity.

316 The net radiation fluxes differed by 20%, and peak net radiation fluxes differed by 16%. In
317 addition, the selection of different plant species also leads to different albedo. Lundholm,
318 Tran [60] reported a wide albedo range (highest value of ~0.26 for *Sedum acre,* and lowest
319 value of ~0.16 for *Rhodiola rasea*), and species in the forbs and grasses group varied slightly
320 between 0.16 ~0.19.

321 *3.1.3 Stomatal resistance*

322 Stomatal resistance reflects the ability of plants to transpire; higher stomatal resistance
323 leads to lower transpiration rates (Table 3 -Eq. 11). The stomata on the leaves are also used
324 to uptake CO_2 from the atmosphere. The stomatal closure depends on the water content in
325 the stomata cavity's interior and the leaf's exterior surface [31]. Plant species with higher
326 stomatal resistance, such as *Sedum*, are normally selected on extensive or semi-extensive
327 roofs because of their ability to tolerate dry conditions [42]. However, succulent plants (e.g.,
328 *Sedum*) produce lower *ET* fluxes than grass plants due to high stomatal resistance, resulting
329 in less cooling [29].

330 *3.2 Substrate*

331 The main functions of a substrate are supporting plant growth, thermal resistance, and
332 heat storage. The substrate is a three-phase system (solid, air, and water in pores). The solid
333 phase of the substrate is composed of aggregates, sand, and organic matter [62]. The substrate
334 comprises lightweight granular solids with lower density and different organic content [14].
335 Albedo, emissivity, thermal conductivity, specific heat capacity, thickness, and water
336 capacity are the most important parameters to evaluate the substrate's thermal mitigation and
337 insulation functions (Fig. 5).

3.2.1 Thermal and spectral properties

As shown in Fig. 5, the thermal conductivity of the GR substrate determines the insulation (Table 3 -Eq. 13) and depends on the substrate composition, i.e., soil particles, density, water content, and organic minerals [63]. The thermal conductivity reported in the literature varies from 0.25 to 0.34 $Wm^{-1}K^{-1}$ for the dry substrate to 0.31 to 0.62 $Wm^{-1}K^{-1}$ for the saturated substrate [64]. GRs offer additional thermal mass to stabilize indoor temperature (Table 3 Eq. 12) [15]. The specific heat capacity of the substrate varies between 0.8 to 1.10 $KJKg^{-1}K^{-1}$ for dry substrate and 1.08 to 1.60 $KJKg^{-1}K^{-1}$ for the saturated substrate [64]. The albedo of the substrate ranges from 0.17 to 0.4 for dry substrate and 0.04 to 0.20 for saturated substrate [64]. However, due to high *LAI* and canopy coverage, substrate albedo does not significantly impact net radiation [61].

3.2.2 Thickness

The thickness of the substrate determines if a GR is extensive (EGR, 5-12 cm) or intensive (IGR, >12 cm) [65]. IGRs usually have higher water capacity and more space for root expansion with a deeper substrate, resulting in higher plant diversity and larger biomass than EGRs [41]. The thicker the substrate, the higher the capacity to sequester carbon in the form of biomass and soil organic carbon (SOC) [66]. Durhman, Rowe [67] showed that the survival and growth rates of plants are higher in deeper substrates because the substrate thickness affects the water and nutrient availability in the substrate. Sun, Bou-Zeid [68] found deeper substrate could store more water, but evaporation did not increase because more water was redistributed into the substrate bottom. The thin substrate of EGRs results in high substrate temperatures and increased evapotranspiration, even water depletion [69]. A thicker substrate reduces conductive fluxes due to higher thermal resistance. Wong, Cheong [70]

found that thermal resistance increased by 0.4 m^2KW^{-1} and 0.0633 m^2KW^{-1} with a 100mm increase in soil thickness for dry and moist clay soils. Stella and Personne [43] found that conductive heat fluxes within the substrate ranged from -20 Wm^{-2} to 35 Wm^{-2} for an EGR but only varied between -10 Wm^{-2} and 5 Wm^{-2} for a semi-IGR in winter. However, the increase of energy saving is less pronounced as the substrate thickness reaches 0.3m [71].

3.2.3 Volumetric water content (VWC)

Volumetric water content (*VWC*) in a substrate is dynamic due to water gains (rainfall, irrigation) and losses (drainage, runoff, evapotranspiration), as shown in Eq. (15). The availability of water has a significant impact on the photosynthesis and respiration. Therefore, it is strongly related to the exchange of CO_2. Heusinger and Weber [72] found the CO_2 uptake declined linearly with *VWC* values ≤ 0.05 m^3 m^{-3}. As a result, *VWC* should exceed 0.05 m^3 m^{-3} to optimize carbon uptake.

Substrate *VWC* is identified as the main factor for evapotranspiration, especially when irrigation is absent (Table 4 Eq. 19-20) [73]. *ET* is restricted due to the lack of water for plant transpiration, increasing the sensible heat flux during the day. Coutts, Daly [74] found that the Bowen ratio is above four during dry daytime but drops below one after irrigation. The evapotranspiration-substrate water content curve presents an elongated "S" shape. *ET* rates are low with scarce water; on the contrary, evapotranspiration rates are high with abundant water availability [31]. High *VWC* lowers substrate temperature due to evaporation. The higher the water retention capacity of the substrate, the larger the cooling potential under warm and dry conditions [75].

VWC is also related to the thermal characteristics of GRs. Albedo decreases with increased substrate moisture, and the thermal conductivity and specific heat capacity of

384 substrate increase at a rate of about 0.038 $Wm^{-1}K^{-1}$ and 211 $Jkg^{-1}K^{-1}$ per 0.1 increase in soil
385 saturation, respectively [64].

386 *3.3 Building environment surrounding of GRs*

387 The thermal performance of GRs is impacted by the surrounding building environment,
388 including building patterns and micro-climate.

389 *3.3.1 Building and urban geometry*

390 Previous studies found that building characteristics (e.g., insulation layer thickness) and
391 urban configuration (e.g., urban canyon geometry, height-to-width ratio, building density)
392 impact the thermal performance of GRs [76, 77].

393 The insulation of the building envelope varies because of physical components such as
394 the roof, walls, doors, windows, etc. As shown in Fig. 5, the building envelope affects the
395 thermal performance of GRs due to heat transfer by the slab and/or insulation layer. He, Yu
396 [78] conducted a sensitivity analysis showing that the insulation effect of GRs was mainly
397 affected by the thermal resistances of the substrate and structure layer. Many studies have
398 found that GRs are more efficient in low-insulation buildings [79].

399 Urban form, i.e., the height, density, and direction of buildings, impacts the thermal
400 performance of GRs by creating distinct microclimates [77]. Sun, Grimmond [80] found the
401 GR cooling benefit to increase linearly with size. Zonato, Martilli [76] showed that tall
402 buildings decrease the cooling effect at ground level and concluded that the impact on 2m air
403 temperature was not linear to building height.

404 *3.3.2 Microclimatic Parameters*

405 As shown in Fig. 5, solar radiation, wind speed, relative humidity, air temperature, and
406 precipitation interact with vegetation, changing the input, transfer, and dissipation of heat and

mass on GRs. Solar radiation is the main energy input on Earth and is the driving force of photosynthesis [81]. Li, Wai [81] found the average CO_2 concentration near plants to be 12.9 mg m^{-3} lower than in the surroundings during the day due to active photosynthesis. In addition, intense solar radiation strongly fuels *ET* [82].

Wind velocity can increase the convective heat fluxes and vapor removal from the substrate. Strong air convection can increase *ET* by quickly replacing saturated air above the surface with drier air [83]. High air temperature and low relative humidity can enhance *ET* [84] (Table 3 Eq. 11) due to a high VPD. Jim and Tsang [85] observed the highest *ET* rate in autumn when relative humidity was lowest. Cascone, Coma [31] recorded higher *ET* rates for the dry substrate on sunny days and similar *ET* rates during cloudy days. Therefore, GRs have larger cooling effects in arid climates [77]. Jamei, Chau [86] reviewed 89 studies in three climates (hot-humid, temperate, and dry) and found the highest median GR cooling effect occurred in dry climates (3℃) and the lowest in hot-humid climates (1℃), showing that the performance of GRs varies with climatic context.

3.4 Maintenance

Management practices to maintain the function and services of GRs include irrigation, weed control, and fertilizing. Irrigation increases water availability in arid conditions to support vegetation growth and increase cooling (Fig. 5). Konopka, Heusinger [18] found an unirrigated GR had the maximum annual uptake of -189gC $m^{-2}y^{-1}$ in a humid year, which roughly doubles the minimum uptake of -95g$m^{-2}y^{-1}$ in a dry year. Results indicate that sustainable irrigation can maintain the carbon uptake function of GRs.

Irrigation changes soil moisture and impacts the *ET* of GRs, which impacts the thermal performance of GRs [52]. However, the higher heat conductivity of substrate with higher VWC could counteract the evaporative cooling effect. Azeñas, Cuxart [87] found lower

conductive heat fluxes in limited-irrigation (25% of PET) treatments than in well-irrigated (50% of PET) ones. In addition, sustainable irrigation allows high-water use plants to survive well apart from drought-enduring plants such as *Sedum*, which altogether can enhance cooling [42].

Weeding is one of the most important and time-consuming tasks in GR maintenance [88]. Weeds compete for nutrients, water, sunlight, and other resources with desired plants. Applying organic fertilizer can directly increase the organic content of the substrate and provide nutrients for plant growth [89]. GR maintenance alters the vegetation and substrate properties, indirectly changing the thermal effects of GRs.

4. Methods to quantify carbon dioxide reduction

In this section, methods for carbon dioxide reduction are summarized by identifying and discussing the hardware and/or software technologies, applications, and spatiotemporal context found in the literature.

4.1 Methods overview

Methods to quantify the carbon dioxide reduction services of GRs can be divided into field measurements and modelling [9] (Table 5). In-situ experiments (e.g., sampling and C/N/S analyzer, allometric equations, Eddy covariance, long-term monitoring system) are usually based on before/after, test/control observations to compare the performance of GRs with bare roofs or other rooftop mitigation strategies (e.g., cool roofs and photovoltaic panels) and to evaluate the performance for different plant species, substrates, and maintenance practices. Experiments are carried out on test plots or full-scale GRs [90]. Observational data show GR performance under real weather conditions, especially during extreme weather

events [45]. Nevertheless, experiments are expensive and time-consuming, and case study results are difficult to extrapolate to different climate conditions. On the other hand, GR simulations (e.g., in EnergyPlus, TRNSYS) can easily be extrapolated to different climate zones and coupled with other models to predict the potential effects of GRs under various scenarios [91]. Modelling studies usually focus on particular countries or climatic zones and assess the thermal performance, energy savings, or irrigation of GRs [14]. Popular models include WRF [76], ENVI-met [92], and life cycle assessment (LCA) techniques [93, 94].

Table 5

The application of field experiment and simulation methods for studying the benefits of GRs. ●= the method can quantify the process or service, ○= can not be used.

Benefits → / Methods ↓	Air CO_2 absorption	Carbon storage	Ambient thermal mitigation	Indoor energy demanding
Field experiment				
Sampling and C/N/S analyzer and/or allometric equations	○	●	○	○
Eddy covariance (EC)	●	○	●	○
Closed chamber	●	○	○	○
Weather station and soil sensors	○	○	●	●
Simulation				
EnergyPlus	○	○	●	●
TRNSYS	○	○	●	●
ENVI-met	○	○	●	●
WRF	○	○	●	●

4.2 Methods to quantify the direct carbon dioxide reduction effect

To assess the direct impacts of GRs on carbon dioxide reduction, studies have measured the storage and fluxes of carbon (CO_2) between the surface and the atmosphere, as well as the conversion of different forms of carbon in different carbon pools in the GR ecosystem. The carbon storage in the GR biomass or substrate is measured using sampling and C/N/S analyzers as well as allometric equations [16]. Whittinghill, Rowe [95] introduced a method

to set quadrats to take random samples of above-ground biomass, below-ground biomass (roots), and substrate at the end of the growing seasons. Allometric equations were developed from destructive sub-sampling to estimate the above-ground and below-ground biomass by the crown volume index (CVI) [95]. Ambient CO_2 concentrations can be quantified using a CO_2/H_2O analyzer or infrared gas analysis (IRGA), and the ability of GRs to remove ambient CO_2 can be assessed by measuring the amount of gas exchanged through leaves under ideal conditions [81, 96, 97].

Closed static chambers and eddy covariance (EC) methods have been used to investigate the greenhouse gas (GHG) fluxes between the atmosphere and GRs [72, 98]. Environmental factors (e.g., solar radiation, air temperature, precipitation, relative humidity, and substrate temperature and moisture) were also recorded in these studies to investigate the net ecosystem CO_2 exchange (NEE) through these environmental factors [18, 99]. The relationship between ecosystem respiration (R_{eco}) and NEE can be measured at test plot scale using the closed chamber method. Eddy-covariance flux measurements can be used to measured exchanges of heat and mass between the GR ecosystem and the overlying atmosphere [100]. Depending on the atmospheric and substrate boundary conditions, the photosynthesis rate and net assimilation of CO_2 can be calculated using a semi-empirical $A - g_s$ model [72, 101].

4.3 Methods to quantify the indirect carbon dioxide reduction effect

GRs improve human thermal comfort and reduce building energy consumption. Methods to quantify indirect dioxide reductions include environmental monitoring and modelling. Weather parameters (solar radiation, wind speed, relative humidity, precipitation, wind direction, and air temperature), substrate moisture and temperature, and heat fluxes

494 have been monitored by various researchers using weather stations and sensors [102, 103].
495 Stella and Personne [43] compared measurements of surface temperatures and conductive
496 heat fluxes reaching the external building envelope to investigate the impact of GRs on
497 building insulation and surface UHI mitigation. Heusinger and Weber [35] used the eddy-
498 covariance (EC) method to quantify the evaporative cooling ability of GRs. Jim [104] used
499 precision energy loggers to monitor and compare air-conditioning electricity consumption of
500 six vacant apartments with varying insulation and GR experimental plots under various
501 weather scenarios.

502 Many models have been developed to simulate the process of heat transfer based on the
503 energy balance, and some models have been coupled with additional mass balance
504 calculations [14]. TRNSYS is a predictive numerical model that calculates the thermal and
505 energy performance of GRs [105]. The main inputs of TRNSYS are the building's geometric
506 elements, meteorological elements (ambient temperature, relative humidity, total radiation,
507 etc.), the building's infiltration, ventilation, etc., while the output is heating and cooling load
508 [91]. A GR model integrated into EnergyPlus proposed by Sailor [56] has also been applied
509 under various scenarios (e.g., different climatic conditions, different irrigation scenarios, and
510 building characteristics) to investigate the energy performances of GRs. EnergyPlus has been
511 used to calculate the insulation effect of GRs. Chan and Chow [106] revised the existing
512 *OTTV* (average heat gain into a building through the building envelope) equations using a
513 validated EnergyPlus simulation and coupled it with the Eco-roof model to calculate the
514 *OTTV* of GRs. He, Lin [107] developed a new model to predict the Roof Thermal Transfer
515 Value (*RTTV*) of GRs based on EnergyPlus results. The long-term energy benefit of GRs has
516 been assessed using Life Cycle Assessment (LCA), and EnergyPlus was used to calculate the
517 energy consumption over the life span of GRs [93, 94, 108]. EnergyPlus was also applied to

simulate the energy performance of GRs under different future climatic conditions over future periods and emission scenarios [109].

The cooling benefits of GRs have been investigated at the neighborhood scale and mesoscale by energy balance models (EBM) and/or computational fluid dynamics models (CFD) [110]. ENVI-met is one of the most widely used CFD models [111] and has been applied to calculate the thermal performance of GRs and energy savings coupled with EnergyPlus [112]. Sun, Grimmond [80] modelled the heat mitigation impact of GRs in the Beijing metropolitan area during heat waves with WRF-PUCM. Zonato, Martilli [76] investigated the building energy consumption in summer and winter in idealized urban areas using WRF. However, these methods have large uncertainties because of model generalizations of processes and factors related to the carbon dioxide reduction of GRs.

5. Outlook and future research needs

The processes, factors, and quantitative methods of recent and important literature were comprehensively reviewed to analyze and evaluate the carbon dioxide reduction performance of GRs in Table 6. Previous studies have shown that GRs are potential carbon sinks, effectively mitigate the urban heat, and improve indoor conditions and energy efficiency of buildings. A gap in understanding the carbon cycling of GR ecosystems was identified. The evaluation of carbon dioxide reductions of GRs is still incomplete and un-robust; thus it requires interdisciplinary approaches. Future scientific research on the carbon dioxide benefits of GRs should focus on three objectives:

(1) A better understanding of GR carbon reduction mechanisms is required to quantify the carbon sequestration benefits of GRs. More studies in various climatic zones are needed

to investigate the interactions among GR elements (i.e., plants, substrate, microbiome) with the environment and their dynamics (e.g., vegetation growth, water, heat transfer).

(2) GR systems customized for specific climate conditions will yield the greatest dioxide reductions. The benefits of GRs vary by geographic region and climate. Determining the right vegetation for local climate conditions is paramount to guaranteeing sustainable GRs. The substrate characteristics should also be studied more in-depth to select the most efficient substrate carefully.

(3) Research at multiple spatial and temporal scales is fundamental to understanding GR ecosystems and evaluating their decarbonization impacts comprehensively. The synergistic effects of GRs on carbon dioxide reductions from the ground level to the city level and from seasonal to annual time scales have not been well studied, and cost-benefit analyses are important to promote and implement GRs. A multi-objective evaluation of the benefits and tradeoffs of GRs is needed to support stakeholder decisions better [113]. For example, irrigation increases evaporative cooling at the cost of more water consumption [68]. Therefore, a precise long-term evaluation and/or prediction of carbon reduction of GRs for multiple spatial and temporal scales is needed in future studies.

Table 6 Representative studies about the processes, critical factors and quantitative methods for direct and indirect carbon dioxide reduction of green roofs.

Carbon dioxide reduction type	Benefits	Processes	Spatial scale	Quantitative methods	Factors	Location	Köppen Climate	Ref.
Direct	Carbon sequestration	Net ecosystem exchange	Experimental modules or building	Closed-chamber, weather station, and soil sensor	Vegetation type, substrate thickness, organic content, and irrigation	Toronto, Canada	Dfb	[114]
					Meteorological, soil parameters, and green roof type	Tartu, Estonia	Cfb	[99]
				Eddy covariance (EC) measurement	Meteorological and soil parameters	South-east of Berlin, Germany	Cfb	[18] [72]
				CO_2/H_2O analyzer and computer simulation	Green roof vs. bare roof	Hong Kong, China	Cwa	[81]
		Carbon storage	Building	Sampling and C/N/S analyzer, and allometric equations	Green roof type	Michigan, America	Dfa	[95]
					Vegetation and substrate	Michigan, and Maryland, America	Dfa	[16]
Indirect	Ambient thermal mitigation	Heat transfer	Building	Weather station	Green roof vs. Bare roof	Nanjing, China	Cfa	[115]
				Eddy covariance (EC) measurement	Meteorological and soil parameters	South-east of Berlin, Germany	Cfb	[35]
			Community-scale	ENVI-met	Meteorological parameters	Hong Kong, China	Cwa	[116]
			City	WRF	Green roof coverage	Beijing, China	Dwa	[80]
			Metropolitan area (16 cities)	WRF	Green roof vs. cool roof	Yangtze River Delta metropolitan	Cfa	[117]
	Ambient thermal mitigation and indoor temperature mitigation	Heat transfer and insulation	Building or Experimental plots	Weather station and soil sensors	EGR and semi-IGR vs. bare roof	Paris, France	Cfb	[43]
					Green roof type	Hong Kong, China	Cwa	[103]
	Ambient thermal mitigation and Energy demanding reduction	Heat transfer and insulation	Building	Weather station and precision energy loggers	Green roof type, weather conditions	Hongkong, China	Cwa	[104]
				EnergyPlus and weather station	Rooftop technologies and climate type	Moroccan	Bsh, Bwh,Csa	[118]
					Vegetation, substrate, and building	Singapore	Af	[119]
					Vegetation structural properties and substrate thickness	Hong Kong, China	Cwa	[106]
				EnergyPlus and LCA	Plant selection and substrate type	Seoul, Republic of Korea	Dwa	[120]
				Soil sensors and TRNSYS	Green roof vs. bare roof	Athens, Greece	Csa	[102]
Direct & indirect	Carbon sequestration and energy demanding reduction	Net ecosystem exchange and heat transfer		Controlled experiment with weather station and plant species observation	Green roof vs. bare roof	Dhaka City, Bangladesh	Aw	[121]
				EnergyPlus, and crop's growth modeling	Building environment	Quito, Ecuador	Cfb	[122]

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